

**Owner-approved ExamSIDE question-demand
slice for plane motion, now semantically: Source
questions**

IIT-JEE · 2011 · Paper II · Q33

Official past paper, faithful restatement of the original

Inside a train accelerating uniformly along a straight track, a boy throws a ball forward at 10 m/s relative to the train and 60 degrees above the horizontal. To catch it again at the launch height he moves 1.15 m forward inside the train. Using $g=10 \text{ m/s}^2$, find the train acceleration in m/s^2 .

Conditions

- Throw speed is relative to the train.
- Train acceleration is constant.
- Catch occurs at the original vertical level.

Answer and working

5 m/s^2 .

1. Vertical motion fixes the common flight time: $T=2(10 \sin 60)/10=\sqrt{3} \text{ s}$.
2. Relative horizontal displacement in the accelerating train is $(10 \cos 60)T - (1/2)aT^2$.

3. Set this equal to 1.15 m: $5\sqrt{3} - 1.5a = 1.15$, giving $a \approx 5 \text{ m/s}^2$.

JEE (Advanced) · 2018 · Paper 2 · Q8

Official past paper, faithful restatement of the original

A ball is launched from level ground at 45 degrees and reaches a maximum height of 120 m. At its first impact with the ground it loses half its kinetic energy. Immediately after the bounce its velocity is at 30 degrees above the horizontal. Find the new maximum height.

Conditions

- First flight starts and ends at ground level.
- Half the kinetic energy is lost at the first bounce.
- Post-bounce launch angle is 30 degrees.

Answer and working

30 m.

1. Initial maximum height gives $u_y^2 = 2g(120)$. At 45 degrees, $u^2 = 2u_y^2 = 480g$.
2. After losing half the kinetic energy, the speed squared becomes $v^2 = u^2/2 = 240g$.

3. The new vertical speed squared is $v^2 \sin^2 30 = 60g$, so $H' = 60g/(2g) = 30$ m.

JEE (Advanced) · 2019 · Paper 2 · Q9

Official past paper, faithful restatement of the original

A projectile is launched from the ground at speed u_0 and angle θ . After each landing it rebounds at the same angle but with speed reduced by a constant factor α : u_0/α , u_0/α^2 , and so on. If the magnitude of average velocity over the entire sequence is 0.8 times the magnitude V_1 of the average velocity over the first flight, find α .

Conditions

- Every rebound uses the same angle θ .
- Successive launch speeds form u_0/α^n .
- The sequence continues indefinitely.

Answer and working

4.

1. For a fixed launch angle, each flight time is proportional to speed and each range is proportional to speed squared.

2. Total displacement is $R1[1+\alpha^{-2}+\alpha^{-4}+\dots]$ while total time is $T1[1+\alpha^{-1}+\alpha^{-2}+\dots]$.
3. Thus $V_{\text{avg}}/V1 = [1/(1-\alpha^{-2})]/[1/(1-\alpha^{-1})] = \alpha/(\alpha+1)=0.8$, giving $\alpha=4$.

JEE (Advanced) · 2021 · Paper 1 · Q5

Official past paper, faithful restatement of the original

A projectile is launched from ground at $5\sqrt{2}$ m/s, at 45 degrees from the vertical. At the highest point it splits into two equal masses. One fragment falls vertically to the ground 0.5 s after splitting. The other fragment reaches the ground t seconds after splitting. Find t .

Conditions

- $g=10 \text{ m/s}^2$.
- The split occurs at the apex.
- Fragments have equal mass.

Answer and working

0.5 s.

1. At the apex the projectile has zero vertical velocity immediately before splitting.
2. The fragment that falls vertically reaches the ground in 0.5 s; this fixes the apex height through $H=(1/2)g(0.5)^2$.

3. Both fragments start from the same height at the same instant; vertical momentum before split is zero, and the vertical-fall fragment has zero horizontal velocity but no stated vertical impulse. The source configuration yields the same vertical fall time for the other fragment, $t=0.5$ s.

JEE (Advanced) · 2021 · Paper 1 · Q6

Official past paper, faithful restatement of the original

Use the same projectile-splitting setup as Q5: launch speed $5\sqrt{2}$ m/s at 45 degrees from vertical; at the apex it splits into equal masses and one fragment drops vertically, reaching ground 0.5 s later. Find the horizontal distance x from the launch point at which the other fragment lands.

Conditions

- $g=10 \text{ m/s}^2$.
- Equal fragments.
- One fragment has zero horizontal velocity immediately after splitting.

Answer and working

7.5 m.

1. The original horizontal speed is 5 m/s and the apex time is 0.5 s, so the split occurs at $x=2.5$ m.
2. Horizontal momentum just before splitting is $M \cdot 5$. If one equal fragment has zero horizontal speed, the other equal fragment must have horizontal speed 10 m/s.

3. During the 0.5 s post-split fall it travels another 5 m horizontally, so $x=2.5+5=7.5$ m.

JEE (Advanced) · 2024 · Paper 2 · Q9

Official past paper, faithful restatement of the original

A ball is thrown from $(0,0)$ with fixed speed v_0 at angle θ_0 . Simultaneously, a stone is thrown from $(L,0)$ toward the ball at angle $180^\circ - \theta_1$ with a suitable speed. For $(\theta_0, \theta_1) = (45^\circ, 45^\circ)$ the hit occurs after T_1 ; for $(60^\circ, 30^\circ)$ it occurs after T_2 . Find $(T_1/T_2)^2$.

Conditions

- Both projectiles are launched simultaneously from the same vertical level.
- Gravity is common to both particles.
- The stone's speed is chosen so that collision occurs.

Answer and working

2.
 1. Because both bodies have the same gravitational acceleration, subtracting their position equations removes the $-gt^2/2$ term.

2. For collision, the vertical relative displacement condition gives the stone speed in terms of v_0 and the two angles; eliminating it yields $T = L \sin(\theta_1) / [v_0 \sin(\theta_0 + \theta_1)]$.
3. Thus $T_1 = L / (\sqrt{2} v_0)$ and $T_2 = L / (2v_0)$, so $(T_1/T_2)^2 = 2$.

IIT-JEE · 2012 · Paper I · Q5

Official past paper, faithful restatement of the original

A block on a frictionless horizontal surface is attached to a spring of natural length 4.9 m. It is stretched by 0.2 m and released, executing SHM with angular frequency $\pi/3$ rad/s. At the same instant a pebble is launched at 45 degrees from a point 10 m away. If it meets the block after 1 s, determine the launch speed. (a) (A) $\sqrt{50}$ m/s (b) (B) $\sqrt{51}$ m/s (c) (C) $\sqrt{52}$ m/s (d) (D) $\sqrt{53}$ m/s

Conditions

- Take $g=10$ m/s².
- The spring-block motion and projectile launch begin simultaneously.
- The block starts from maximum extension.

Answer and working

Option (A): $\sqrt{50}$ m/s.

1. After 1 s the spring extension is $0.2 \cos(\pi/3)=0.1$ m, so the block is 5.0 m from the fixed end.
2. The pebble starts 10 m away and must cover 5 m horizontally in 1 s, so $v \cos 45=5$.

3. Hence $v=5\sqrt{2}=\sqrt{50}$ m/s; its vertical displacement after 1 s is also zero, so the meeting height is consistent.

JEE (Advanced) · 2022 · Paper 1 · Q8

Official past paper, faithful restatement of the original

A projectile has ordinary range d when launched at speed v and angle θ under gravity g . At the highest point it enters a region where the effective downward acceleration is $g/0.81$. If its new total horizontal range is $d' = n d$, find n .

Conditions

- The acceleration changes only after the projectile reaches its apex.
- Horizontal velocity is unchanged at the boundary.

Answer and working

0.95.

1. The projectile reaches the apex after covering half of its ordinary range, $d/2$.
2. The apex height is unchanged, but the downward acceleration becomes $g/0.81$, so the fall time from the apex is 0.9 times the original descent time.

3. The second horizontal segment is therefore $0.9(d/2)$; hence $d' = d/2 + 0.9d/2 = 0.95d$.

JEE (Advanced) · 2023 · Paper 1 · Q1

Official past paper, faithful restatement of the original

A ball slides without friction from a height h above a terrace whose top is $3h$ above the ground, leaves horizontally, strikes the ground and rebounds with coefficient of restitution $1/\sqrt{3}$. Determine which of the source statements about launch speed, impact angle, rebound velocity and rebound geometry are correct. (a) (A) $u_0 = \sqrt{2gh}$ horizontally (b) (B) rebound velocity equals $\sqrt{2gh}(\hat{x} - \hat{z})$ (c) (C) impact angle is 60° (d) (D) horizontal range divided by rebound maximum height is $2\sqrt{3}$

Conditions

- Slide is frictionless.
- Ground is horizontal.
- Coefficient of restitution is $1/\sqrt{3}$.

Answer and working

Options (A), (C) and (D).

1. Energy on the slide gives horizontal launch speed $u_0 = \sqrt{2gh}$.

2. Falling through $3h$ gives vertical impact speed $\sqrt{6gh}$, so $\tan(\theta) = \sqrt{6gh} / \sqrt{2gh} = \sqrt{3}$, hence $\theta = 60$ degrees.
3. After impact the upward vertical component is $\sqrt{2gh}$; rebound height is h , while pre-impact flight gives horizontal distance $2\sqrt{3}h$, so $d/h = 2\sqrt{3}$.

JEE (Advanced) · 2025 · Paper 2 · Q15

Official past paper, faithful restatement of the original

A 0.2 kg projectile is launched at 270 m/s at 60 degrees in a medium with linear drag $F = -c v$, where $c = 0.1$ kg/s. It hits a vertical wall after 2 s. Taking $e = 2.7$ for the exponential evaluation specified by the source, determine the horizontal distance to the wall.

Conditions

- Linear viscous drag acts opposite the instantaneous velocity.
- Use the source's $e = 2.7$ approximation.

Answer and working

170 m (within the official accepted interval 167–171 m).

1. The horizontal component obeys $m \, dv_x/dt = -c \, v_x$, so $v_x(t) = v_{x0} \exp[-(c/m)t]$.
2. Here $v_{x0} = 270 \cos 60 = 135$ m/s and $m/c = 2$ s, so $x(t) = v_{x0}(m/c)[1 - \exp(-ct/m)]$.

3. At $t=2$ s, $x=270(1-1/e)=270(1-1/2.7)=170$ m.

IIT-JEE · 2011 · Paper II · Q26

Official past paper, faithful restatement of the original

A 0.2 kg ball rests on top of a 5 m vertical post. A 0.01 kg bullet moving horizontally with speed V strikes the ball centrally. After the collision the ball and bullet move independently and land 20 m and 100 m, respectively, from the foot of the post. Find the bullet's initial speed. (a) (A) 250 m/s (b) (B) $250\sqrt{2}$ m/s (c) (C) 400 m/s (d) (D) 500 m/s

Conditions

- Both bodies begin their post-collision projectile motion from the same 5 m height.
- Use $g=10 \text{ m/s}^2$ for the numerical evaluation consistent with the source options.

Answer and working

Option (D): 500 m/s.

1. From height 5 m with zero initial vertical velocity, the common fall time is $t=\sqrt{2h/g}=1 \text{ s}$.
2. Therefore the ball and bullet post-collision horizontal speeds are 20 m/s and 100 m/s.

3. Horizontal momentum during the short collision is conserved: $0.01V = 0.2(20) + 0.01(100) = 5$, giving $V = 500$ m/s.

JEE (Advanced) · 2014 · Paper 1 · Q11

Official past paper, faithful restatement of the original

Airplanes A and B fly at constant velocities in the same vertical plane, at 30 degrees and 60 degrees to the horizontal. A moves at $100\sqrt{3}$ m/s. At $t=0$, B is 500 m from A, and an observer in A sees B move perpendicular to A's flight direction. If A just avoids being hit by B, find the encounter time t_0 .

Conditions

- Both airplanes move with constant velocity.
- Their trajectories are straight lines in one vertical plane.
- Initial separation is 500 m.

Answer and working

5 s.

1. Resolve A's velocity as $(150, 50\sqrt{3})$ m/s. Let B's speed be u , so its velocity is $(u/2, u\sqrt{3}/2)$.
2. The relative velocity $v_{B/A}$ is perpendicular to A's velocity. Their dot product is zero, which gives $u=200$ m/s.

3. Then $v_{B/A}=(-50,50\sqrt{3})$ m/s, whose magnitude is 100 m/s. Closing 500 m at that relative speed gives $t_0=5$ s.

JEE (Advanced) · 2026 · Paper 1 · Q6

Official past paper, faithful restatement of the original

A projectile launched from O with speed v at angle θ passes through P, which is 5 m horizontally and 1 m vertically from O. Determine which source statements are correct about the required speed and whether the projectile reaches its maximum height before or after P for specified launch angles. (a) (A) For $\theta=45^\circ$, $v=5\sqrt{g}/2$ m/s (b) (B) For $\theta=45^\circ$, maximum height is reached before P (c) (C) For $\theta=30^\circ$, maximum height is reached after P (d) (D) For $\theta=\tan^{-1}(1/5)$, $v=125\sqrt{g}$ m/s

Conditions

- Uniform gravitational acceleration g acts downward.
- Point P has coordinates (5 m, 1 m) relative to launch point O.

Answer and working

Options (A) and (B).

1. Use $y=x \tan(\theta)-g x^2/[2v^2 \cos^2(\theta)]$ at $P=(5,1)$. For $\theta=45^\circ$, this gives $v^2=25g/4$, so $v=5\sqrt{g}/2$; (A) is true.

2. For the same 45° case, the trajectory slope at $x=5$ is negative, so the apex was passed before P; (B) is true.
3. For $\theta=30^\circ$ the apex is not after P, and $\theta=\tan^{-1}(1/5)$ would make the straight-line term equal exactly to $y=1$ before gravity lowers the path, so no finite v can satisfy (D).

JEE (Advanced) · 2013 · Paper 1 · Q7

Official past paper, faithful restatement of the original

A particle of mass m is projected from the ground with speed u_0 at angle α . At the highest point of its trajectory it undergoes a completely inelastic collision with an identical particle that has been thrown vertically upward from the ground with the same initial speed u_0 . Find the angle made by the composite body with the horizontal immediately after collision. (a) (A) $\pi/4$ (b) (B) $\pi/4 + \alpha$ (c) (C) $\pi/2 - \alpha$ (d) (D) $\pi/2$

Conditions

- The first collision occurs at the highest point of the oblique projectile.
- The second particle has the same mass and the same launch-speed magnitude u_0 .
- The collision is completely inelastic.

Answer and working

Option (A): $\pi/4$.

1. At the first particle's apex, its speed is purely horizontal with magnitude $u_0 \cos(\alpha)$.

2. The collision height is $H = \frac{u_0^2 \sin^2(\alpha)}{2g}$. A second identical particle launched vertically upward from the ground with speed u_0 has speed $\sqrt{u_0^2 - 2gH} = u_0 \cos(\alpha)$ at that height.
3. The two equal masses therefore bring equal-magnitude perpendicular momenta into a completely inelastic collision, so the composite momentum bisects the right angle and makes angle $\pi/4$ with the horizontal.

JEE Main · 2026 · Session 2 · 04 Apr 2026 · Shift 2 · Q29

Official past paper, faithful restatement of the original

A projectile has coordinates $x=24t$ and $y=43.6t-4.9t^2$, with x and y in metres and t in seconds. What angle does its velocity vector make with the horizontal at $t=2$ s? (a) 60° (b) 45° (c) 30° (d) 75°

Answer and working

45° .

1. Differentiate the coordinate laws: $v_x=dx/dt=24$ m/s and $v_y=dy/dt=43.6-9.8t$.
2. At $t=2$ s, $v_y=43.6-19.6=24$ m/s.
3. Thus $\tan(\theta)=v_y/v_x=1$, so $\theta=45^\circ$.